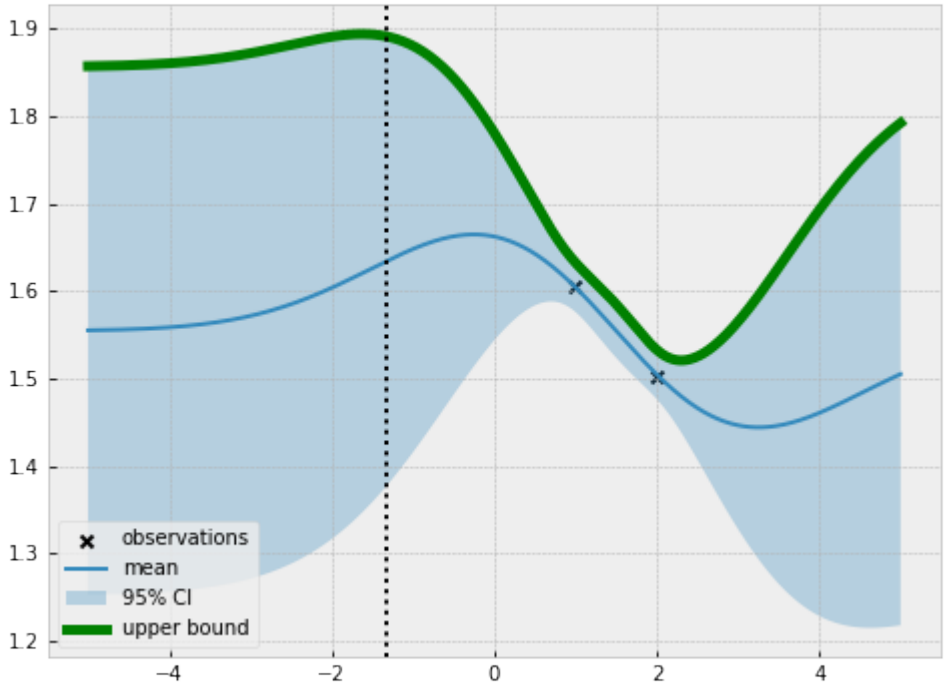


Figure 5.6 The Upper Confidence Bound of a Gaussian process that corresponds to the 95% credible interval. This bound can be used as the acquisition score of the Upper Confidence Bound policy.

SIDEBAR **Acquisition scores**

The acquisition scores quantify how valuable a data point is in guiding us towards the optimum of the objective function. We first learned about acquisition scores in Section 4.1.1 of Chapter 4.

You might think that this way of decision making might not be appropriate, especially in high-stakes situations where the cost of a bad decision is high. In our umbrella example, by optimistically underestimating the probability of rain, you might run the risk of getting caught in the rain without an umbrella.

However, it is a particularly efficient way of reasoning, as we simply need to extract an upper bound of our estimate of a quantity of interest, and use that bound for decision making. Moreover, as we will see in the next subsection, by choosing which CI we'd like to use (as opposed to sticking with the 95% CI), we have complete control over how optimistic UCB is. This control also allows the policy to balance between exploration and exploitation, which is the core question that needs to be addressed by any BayesOpt policy.

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